Lecture 15

Complex arithmetic, roots and rotations

Uses.

Summary 2: Composition of maps and induction are defined. Standing fact E supplies the trigonometric identities and their usual periods, and positive real powers and their laws; Lecture 4 uses their positive nth roots. The uniqueness needed here is justified below.
Summary 12: The Euclidean length in $\mathbb{R}^2$ is $\sqrt{x^2+y^2}$ and obeys the triangle inequality.
Summary 14: Polar coordinates represent a nonzero planar point uniquely with positive radius and azimuth in $[0, 2\pi)$. Other arguments differ by full turns.

Objectives. *A multiplication on the coordinate plane produces a number whose square is* -1. *Moduli express distance conditions in that plane. The geometric meaning of multiplication by a nonzero number is rotation combined with a positive change of scale. Polar coordinates then turn the equation* $w^n = z$ *into a length equation and an angle equation. A local two-by-two matrix calculation records the same multiplication for later linear algebra.*

By the end of this lecture you will be able to:

- Solve complex arithmetic and modulus equations, dividing only by nonzero denominators.
- Prove the polar multiplication law and De Moivre's formula for integer powers.
- Find all nth roots and prove the roots-of-unity sum, including $n = 1$.
- Verify the real two-by-two matrix representation of complex multiplication.

15.1 A multiplication on the plane

The set of *complex numbers*, denoted by $\mathbb{C}$, consists of ordered real pairs. We write the pair (a, b) as $a + \mathrm{i}b$. Its *real part* is $\operatorname{Re} z = a$ and its *imaginary part* is $\operatorname{Im} z = b$. The imaginary part is itself a real number. Equality means equality of both parts. The arithmetic operations are defined by

$$(a + \mathrm{i}b) + (c + \mathrm{i}d) = (a + c) + \mathrm{i}(b + d), \qquad (a + \mathrm{i}b)(c + \mathrm{i}d) = (ac - bd) + \mathrm{i}(ad + bc).$$

The real number a is identified with the pair $(a, 0)$. The *imaginary unit* is $\mathrm{i} = (0, 1)$. The multiplication rule gives $\mathrm{i}^2 = -1$. The additive and multiplicative identities are 0 and 1 under the real identification, so ordinary real arithmetic is preserved.

Addition is commutative and associative because it is componentwise. The product is commutative: exchanging the pairs leaves $ac - bd$ and $ad + bc$ unchanged. Distributivity

follows by distributing each real component over a sum. For associativity, let $z = a + \mathrm{i}b$, $w = c + \mathrm{i}d$ and $u = p + \mathrm{i}q$. Expanding the defining rule in either order gives the same parts:

$$\operatorname{Re}((zw)u) = \operatorname{Re}(z(wu)) = acp - bdp - adq - bcq,$$

$$\operatorname{Im}((zw)u) = \operatorname{Im}(z(wu)) = acq - bdq + adp + bcp.$$

Thus products may be grouped without changing their value. The usual distribution and collection of powers of i now follows from these proved arithmetic rules and $\mathrm{i}^2 = -1$.

Example 15.1 (Multiplication in Cartesian form)**.** Compute the product of $3 + 4\mathrm{i}$ and $-2 + \mathrm{i}$. Distribution gives

$$(3 + 4\mathrm{i})(-2 + \mathrm{i}) = -6 + 3\mathrm{i} - 8\mathrm{i} + 4\mathrm{i}^2 = -10 - 5\mathrm{i}.$$

Its real and imaginary parts are -10 and -5.

15.2 Conjugation, modulus and division

For $z = a + \mathrm{i}b$, its *complex conjugate* and *modulus* are

$$\overline{z} = a - \mathrm{i}b, \qquad |z| = \sqrt{a^2 + b^2}.$$

Conjugation reflects the corresponding point across the real axis. The modulus is its Euclidean distance from the origin. It is nonnegative and vanishes exactly at $z = 0$. For a real number it agrees with the real absolute value. The product and sum satisfy

$$z\overline{z} = |z|^2, \qquad \overline{z + w} = \overline{z} + \overline{w}, \qquad \overline{zw} = \overline{z}\,\overline{w}.$$

The first identity follows from $(a + \mathrm{i}b)(a - \mathrm{i}b) = a^2 + b^2$. The sum identity negates both imaginary components. For the product identity, both sides equal $(ac - bd) - \mathrm{i}(ad + bc)$.

The modulus is multiplicative. Indeed, expanding the real squares gives

$$|zw|^2 = (ac - bd)^2 + (ad + bc)^2 = (a^2 + b^2)(c^2 + d^2) = |z|^2|w|^2.$$

Taking nonnegative square roots proves $|zw| = |z||w|$. Lecture 12's vector triangle inequality, applied to the same ordered pairs, gives $|z + w| \le |z| + |w|$.

If $z \neq 0$, then $|z|^2 > 0$ and

$$z^{-1} = \frac{\overline{z}}{|z|^2}. \tag{15.1}$$

Multiplication by z gives 1 by the first conjugation identity. This reciprocal is unique: if $zv = 1$, multiplication by the displayed reciprocal and associativity give $v = z^{-1}$. Division by a nonzero number is multiplication by its reciprocal. For $z \neq 0$ and an integer $m \ge 1$, define the negative power by $z^{-m} = (z^{-1})^m$. For $w \neq 0$, the modulus law then gives $|z/w| = |z|/|w|$. Consequently, a product of two nonzero complex numbers cannot be zero.

Example 15.2 (A quotient)**.** Compute $(2 - 3\mathrm{i})/(1 + 2\mathrm{i})$. Multiplying numerator and denominator by the conjugate of the denominator gives

$$\frac{2 - 3\mathrm{i}}{1 + 2\mathrm{i}} = \frac{(2 - 3\mathrm{i})(1 - 2\mathrm{i})}{1^2 + 2^2} = \frac{-4 - 7\mathrm{i}}{5}.$$

The denominator is nonzero since its modulus is $\sqrt{5}$. Multiplication checks the answer: $(1 + 2\mathrm{i})(-4/5 - 7\mathrm{i}/5) = 2 - 3\mathrm{i}$. Its modulus is $\sqrt{13}/\sqrt{5}$, in agreement with the quotient law.

15.3 Distance equations in the complex plane

For fixed $p \in \mathbb{C}$, the modulus $|z - p|$ is the distance from p to z. The equation $|z - p| = R$, with $R > 0$, is therefore a circle of centre p and radius R. With $R = 0$ it describes the single point p. A distance equation can be converted to a real equation by squaring its nonnegative sides. From the conjugation rules,

$$|z - p|^2 = (z - p)(\overline{z} - \overline{p}) = |z|^2 - z\overline{p} - p\overline{z} + |p|^2 = |z|^2 - 2\operatorname{Re}(z\overline{p}) + |p|^2.$$

The last step uses $p\overline{z} = \overline{z\overline{p}}$ and the identity $w + \overline{w} = 2\operatorname{Re} w$. For distinct fixed points p and q, the equation of equal distances is consequently

$$|z - p| = |z - q| \quad \Longleftrightarrow \quad 2\operatorname{Re}\big(z(\overline{q} - \overline{p})\big) = |q|^2 - |p|^2.$$

To read it geometrically, observe that $\operatorname{Re}((x + \mathrm{i}y)\overline{a + \mathrm{i}b}) = ax + by$, the planar dot product. The displayed equation has nonzero normal direction $q - p$. The midpoint $(p + q)/2$ satisfies the equal-distance condition. Thus its set is the line through that midpoint perpendicular to the segment from p to q, called the *perpendicular bisector*.

Figure 15.1 shows a perpendicular bisector together with a circle. Their common points satisfy both distance conditions in the example.

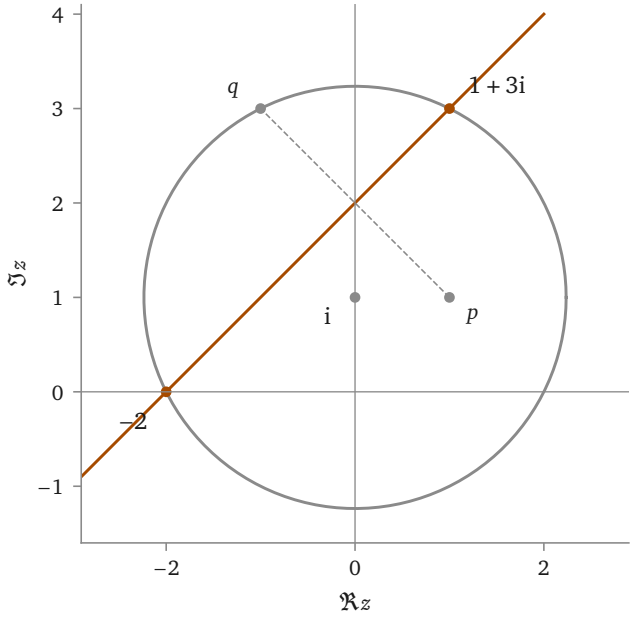


Figure 15.1: The perpendicular bisector for $p = 1 + \mathrm{i}$ and $q = -1 + 3\mathrm{i}$ meets the circle of centre i and radius $\sqrt{5}$ at the two marked common points.

Example 15.3 (Two distance conditions)**.** Find every $z \in \mathbb{C}$ satisfying

$$|z - (1 + \mathrm{i})| = |z - (-1 + 3\mathrm{i})|, \qquad |z - \mathrm{i}| = \sqrt{5}.$$

Write $z = x + \mathrm{i}y$. Squaring the first condition gives $(x - 1)^2 + (y - 1)^2 = (x + 1)^2 + (y - 3)^2$. Expanding and cancelling gives $-4x + 4y - 8 = 0$, hence $y = x + 2$. The squared circle condition is $x^2 + (y - 1)^2 = 5$. Substitution gives

$$x^2 + (x + 1)^2 = 5, \qquad x^2 + x - 2 = (x - 1)(x + 2) = 0.$$

The candidates are $(x, y) = (1, 3)$ and $(-2, 0)$, giving $z = 1 + 3\mathrm{i}$ and $z = -2$. For the first candidate the equal distances are both 2. For the second they are both $\sqrt{10}$. Both have distance $\sqrt{5}$ from i. They therefore satisfy the original unsquared conditions, and the preceding equivalences exclude any other points.

15.4 Polar multiplication and rotations

For $z \neq 0$, write its planar polar representation as

$$z = r(\cos\varphi + \mathrm{i}\sin\varphi), \qquad r = |z| > 0.$$

An *argument* of z is any real angle φ giving this expression. Arguments differ by $2\pi k$ with $k \in \mathbb{Z}$. We denote the unique representative in $[0, 2\pi)$ by $\operatorname{Arg} z$. The zero complex number has modulus zero and no argument. For convenience, define $\operatorname{cis}\varphi = \cos\varphi + \mathrm{i}\sin\varphi$. Its modulus is 1, and the addition identities from E give

$$\operatorname{cis}\varphi \operatorname{cis}\psi = (\cos\varphi\cos\psi - \sin\varphi\sin\psi) + \mathrm{i}(\sin\varphi\cos\psi + \cos\varphi\sin\psi) = \operatorname{cis}(\varphi + \psi).$$

Consequently, multiplication of nonzero complex numbers multiplies their moduli and adds their arguments, with the resulting argument understood modulo 2π.

Fix a nonzero number $w = R\operatorname{cis}\psi$. The map $z \mapsto wz$ rotates every nonzero position by angle ψ and multiplies its distance from the origin by $R > 0$. It fixes the origin. Multiplication by zero instead sends every position to the origin. For the earlier multiplier $w = 3 + 4\mathrm{i}$, the scale factor is 5 and the rotation angle is $\arctan(4/3)$, since both its parts are positive. The computed image of $-2 + \mathrm{i}$ is therefore a rotated position with five times its original length.

Multiplication by i has Cartesian formula $\mathrm{i}(a + \mathrm{i}b) = -b + \mathrm{i}a$. Figure 15.2 illustrates this quarter-turn.

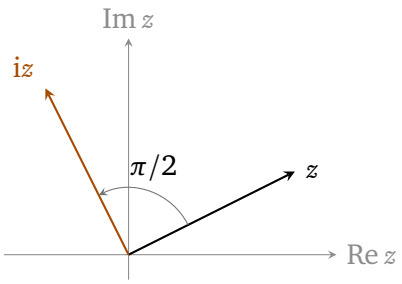


Figure 15.2: Multiplication by i turns every planar vector through $\pi/2$ counterclockwise and preserves its length.

Theorem 15.4 (De Moivre's formula). *For every $\varphi \in \mathbb{R}$ and integer n,*

$$(\cos\varphi + \mathrm{i}\sin\varphi)^n = \cos(n\varphi) + \mathrm{i}\sin(n\varphi).$$

Proof. For $n = 0$ both sides equal 1. For $n \geq 1$, induction follows from the polar multiplication law: multiplying the order-n formula by $\operatorname{cis}\varphi$ adds one more φ to its argument. Also $\operatorname{cis}\varphi \operatorname{cis}(-\varphi) = 1$. For $n = -m$ with $m \geq 1$, taking the mth power of this reciprocal gives $\operatorname{cis}(-m\varphi)$ by the positive case. The factor has modulus 1 and is never zero, so all these negative powers are defined. ∎

15.5 All roots of a complex number

Let $n \geq 1$ be an integer. An *nth root* of z is a solution $w \in \mathbb{C}$ of $w^n = z$. If $z = 0$, taking moduli gives $|w|^n = 0$, so $w = 0$ is the only solution. For nonzero $z = \rho \operatorname{cis} \varphi$, every solution is nonzero. Write it as $w = r \operatorname{cis} \theta$ with $r > 0$. Taking moduli and then arguments gives

$$r^n = \rho, \qquad n\theta = \varphi + 2\pi k \quad (k \in \mathbb{Z}).$$

Standing fact E supplies the positive solution $r = \rho^{1/n}$. It is unique. For $n = 1$ this is immediate; for $n \geq 2$, if $0 < r < s$, then $s^n - r^n = (s - r)(s^{n-1} + s^{n-2}r + \cdots + r^{n-1}) > 0$. Thus two distinct positive numbers cannot have the same nth power. The second and De Moivre's formula give exactly the following roots:

$$w_k = \rho^{1/n} \operatorname{cis}\left(\frac{\varphi + 2\pi k}{n}\right), \qquad k = 0, 1, \ldots, n - 1. \tag{15.2}$$

They are distinct. Equality for two indices would make their angle difference $2\pi(k - j)/n$ a whole turn, forcing $k = j$ in this index range. Every integer index differs from one in the range by a multiple of n, which only adds a whole turn to the root's argument. Thus no solutions are missing. Changing the argument representative of z permutes the same list. These roots are equally spaced on their radius circle. They form the vertices of a regular n-gon when $n \geq 3$; for $n = 2$ they are an opposite pair. The notation $\rho^{1/n}$ in the radius is a positive real root. For a complex z, a single-valued notation $z^{1/n}$ would require a further choice from the displayed list.

Figure 15.3 shows the roots used in the next example.

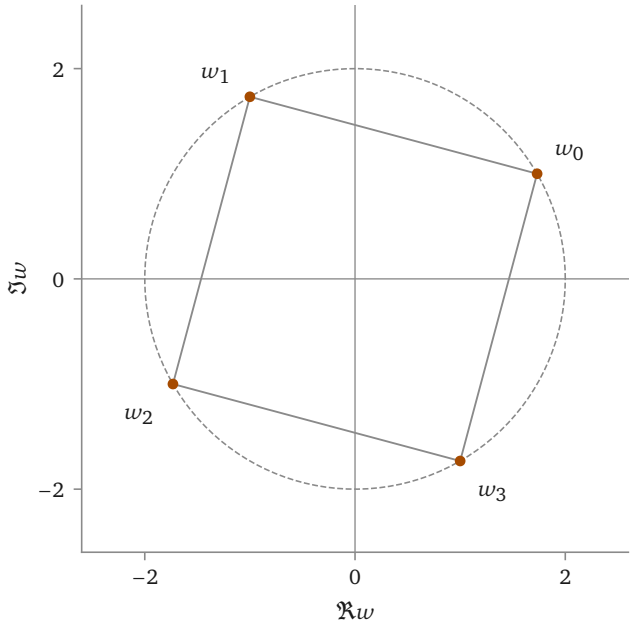


Figure 15.3: The four fourth roots of $-8 + 8\sqrt{3}\,\mathrm{i}$ have radius 2 and successive argument difference $\pi/2$.

Example 15.5 (Fourth roots)**.** Find all solutions of $w^4 = -8 + 8\sqrt{3}\,\mathrm{i}$. The right side has modulus 16 and argument $2\pi/3$. The roots have modulus 2 and arguments $\pi/6 + k\pi/2$, for $k = 0, 1, 2, 3$. They are

$$w_0 = \sqrt{3} + \mathrm{i}, \quad w_1 = -1 + \sqrt{3}\,\mathrm{i}, \quad w_2 = -\sqrt{3} - \mathrm{i}, \quad w_3 = 1 - \sqrt{3}\,\mathrm{i}.$$

For a direct check, $w_0^2 = 2+2\sqrt{3}\,\mathrm{i}$, whose square is $-8+8\sqrt{3}\,\mathrm{i}$. Each successive root multiplies w_0 by another power of i, whose fourth power is 1. The four listed roots therefore solve the equation, and the general formula proves that there are no others.

An nth *root of unity* solves $w^n = 1$. Put $\zeta = \mathrm{cis}(2\pi/n)$. The roots are $1, \zeta, \ldots, \zeta^{n-1}$. For $n > 1$, the root ζ differs from 1. For any complex q, put $S = 1 + q + \cdots + q^{n-1}$. Associativity and distributivity give $(1-q)S = 1 - q^n$ by cancellation. When $q \neq 1$, division by $1 - q$ gives the finite geometric identity; this is the same cancellation as on Sheet 3, now applied to complex arithmetic. Hence

$$(1-\zeta)(1+\zeta+\cdots+\zeta^{n-1}) = 1 - \zeta^n = 0.$$

Division by $1 - \zeta$ shows that their sum is zero. For $n = 1$ the sole root is 1 and its sum is 1. In the fourth-root example, the four coordinate pairs also sum to zero, as predicted by multiplying the four roots of unity by w_0.

15.6 A real matrix representation

A real *two-by-two matrix* is an array with two rows and two columns. For this section, define addition and real scalar multiplication entrywise, and define multiplication by

$$\begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} u & v \\ w & t \end{pmatrix} = \begin{pmatrix} pu + qw & pv + qt \\ ru + sw & rv + st \end{pmatrix}.$$

These local definitions will be generalised in Lecture 19. The matrix acts on an ordered pair (x, y) by giving the pair $(px + qy, rx + sy)$. Distributing the expressions shows that the displayed matrix product acts by first applying the right matrix and then the left. Two matrices with the same action on every pair are equal, since their actions on $(1, 0)$ and $(0, 1)$ recover their columns. For three matrices, either grouping of their product acts on a pair by applying the third, then the second, then the first matrix. Associativity of function composition therefore proves associativity of this matrix product.

Define the map Φ from $\mathbb{C}$ to the set of real two-by-two matrices by

$$\Phi(a + \mathrm{i}b) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}.$$

Its action on the pair corresponding to $x + \mathrm{i}y$ gives $(ax - by, bx + ay)$, exactly the pair for $(a+\mathrm{i}b)(x+\mathrm{i}y)$. The first column recovers a and b, so the map is injective. Entrywise addition gives $\Phi(z + w) = \Phi(z) + \Phi(w)$. For $z = a + \mathrm{i}b$ and $w = c + \mathrm{i}d$, the local product gives

$$\Phi(z)\Phi(w) = \begin{pmatrix} ac - bd & -(ad + bc) \\ ad + bc & ac - bd \end{pmatrix} = \Phi(zw).$$

Thus the map preserves multiplication as well. The identity and imaginary unit become

$$\Phi(1) = I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \Phi(\mathrm{i}) = J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The matrix I_2 leaves every pair unchanged. For a real two-by-two matrix A and an integer $n \geq 1$, the power A^n is the repeated product of n copies of A. Associativity makes the grouping irrelevant. We set $A^0 = I_2$. The matrix J gives the same quarter-turn as multiplication by i, and $J^2 = -I_2$ follows from the proved product law. This representation relates complex arithmetic to composition of real coordinate transformations. It uses no determinant or matrix-inversion theorem.

Summary 15.

- Complex numbers are real pairs with the stated arithmetic. Conjugation gives $z\overline{z} = |z|^2$, and every nonzero number has reciprocal (15.1). Moduli multiply and obey the triangle inequality.
- Polar multiplication adds arguments and multiplies moduli. Theorem 15.4: De Moivre's formula holds for every integer exponent. Arguments of a nonzero number are defined modulo 2π.
- Equation (15.2): a nonzero number has exactly n distinct nth roots for an integer $n \geq 1$. Zero has only the root zero. Roots of unity sum to zero for $n > 1$, and to 1 for $n = 1$.
- § 15.6: the injective map Φ preserves sums and products under the defined two-by-two matrix operations.

Sheet 15 uses these methods in arithmetic, root equations and coordinate transformations. Lecture 16 introduces complex exponential notation and factorisation of real polynomials.

References

K. F. Riley, M. P. Hobson and S. J. Bence, *Mathematical Methods for Physics and Engineering*, 3rd ed., Cambridge University Press, 2006, Chapter 3.